

ICIPS ANALYTICS

INDIAN CRIME INTELLIGENCE & PUBLIC SAFETY ANALYTICS

FULL PROJECT REPORT

Data-driven insights for safer communities

Project focus: Transform reported crime records into structured, interpretable and decision-support information through data cleaning, classification, analytics, visualization and risk-oriented analysis.

Data • Analytics • Visualization • Public-Safety Insights

1. Abstract

Indian Crime Intelligence & Public Safety Analytics (ICIPS) is a data analytics project designed to transform crime records into structured, interpretable and decision-support information. The project combines data understanding, cleaning, classification, statistical aggregation and interactive visualization into a stakeholder-oriented analytical dashboard.

The system analyzes reported crime by city, crime domain, crime category, victim demographics, weapons, case status and time. The current web application is a client-side dashboard implemented with HTML5, CSS3, Vanilla JavaScript and ApexCharts. Python and SQL/SQLite are defined as supporting layers for cleaning, extended analytics and normalized analytical storage.

The project is organized around five stakeholder perspectives: Executive Overview, Police Operations, Crime Intelligence, Women's Safety and City Risk. A dynamic insight layer is intended to translate filtered analytical outputs into concise observations.

The supplied dashboard displays 40,160 reported crimes, 20,062 closed cases, 20,098 open cases, a 50.0% case-closure rate and a 28.6% violent-crime share. Delhi is shown as the highest-volume city in the displayed comparison, while Theft is the highest reported crime category. These values are treated as dashboard-displayed project results, not independently revalidated official statistics.

Keywords: crime analytics, data visualization, public safety, dashboard, KPI, SQL, SQLite, JavaScript, Python, risk analysis, city intelligence.

2. Introduction

Crime datasets can contain a large number of individual records, but raw records are difficult to use directly for comparison, trend detection and decision support. ICIPS addresses this gap by turning record-level information into reusable dimensions, aggregated measures, visualizations and concise insights.

The project follows a raw-data-to-intelligence workflow: load the source records, validate and clean values, derive analytical attributes, organize the data for analysis, calculate KPIs, visualize patterns and generate filter-responsive insights.

The current interface is designed as a single-page analytical application with a premium dark visual language, sticky navigation, responsive layouts, interactive charts and loading/error states. The project documentation explicitly distinguishes currently implemented dashboard capabilities from proposed database, risk-analysis and future-integration layers.

Project identity

Item	Description
Project	Indian Crime Intelligence & Public Safety Analytics (ICIPS)
Domain	Data Analytics / Crime Intelligence / Public Safety
Primary Data	Crime records containing report, occurrence, victim, weapon, city and case-status fields
Dashboard	Interactive client-side analytics dashboard
Core Technologies	HTML5, CSS3, Vanilla JavaScript, ApexCharts, Python, SQL/SQLite design

3. Project Overview

ICIPS is organized as a layered analytical product. Each layer has a defined responsibility so that data preparation, analytics and presentation remain conceptually separate.

Layer	Purpose	Technology
Data	Raw crime records	JSON / dataset files
Processing	Cleaning, parsing and classification	JavaScript / Python
Database	Structured relational storage	SQL / SQLite design
Analytics	Aggregations, trends and distributions	SQL + Python
Visualization	KPIs and interactive charts	ApexCharts + Web UI
Presentation	Stakeholder-facing dashboard	HTML5 + CSS3 + JavaScript

Stakeholder perspectives

- Executive Overview — high-level crime volume, case outcomes and public-safety patterns.
- Police Operations — operational workload, deployment, backlog and city/category comparisons.
- Crime Intelligence — crime types, temporal patterns, victim demographics and weapons.
- Women's Safety — female-victim-oriented demographic and case-outcome analysis.
- City Risk — city comparisons, rankings, compositions and risk-oriented analysis.

A dynamic insight generator is also described so observations can react to active filters rather than remaining fixed narrative text.

4. Problem Statement

A flat crime-event dataset can answer individual record questions but does not automatically provide an analytical view of city-level concentration, dominant categories, case resolution, demographic patterns or temporal changes. Without structured transformation and aggregation, important patterns can remain difficult to interpret.

ICIPS addresses the problem by converting individual records into reusable analytical dimensions, aggregated facts, KPIs, charts and decision-support insights.

Key questions addressed

- Which cities have the highest number of reported crimes?
- Which crime categories and domains are most frequently reported?
- How does reported crime change over time?
- What proportion of cases are closed versus open?
- What demographic patterns appear in reported cases?
- Which locations or city contexts show higher concentrations?
- How can derived risk indicators be used responsibly?

5. Objectives

- Create a structured analytical foundation for crime-event data.
- Clean and standardize dates, categories, demographics and case-status fields.
- Classify crime descriptions into useful crime domains and categories.
- Analyze crime by city, date, month, time, category, weapon and victim attributes.
- Measure open and closed cases and calculate case-closure rates.
- Identify high-volume cities and commonly reported crime categories.
- Provide dashboard-ready data and visualizations for multiple stakeholders.
- Support location-oriented risk analysis as a derived analytical layer.
- Maintain a transparent distinction between source fields and calculated indicators.
- Provide a foundation for future live data integration and predictive analytics.

Expected outcome

The expected outcome is an interactive crime-intelligence dashboard that makes complex crime data easier to explore and converts statistical results into concise visual insights for decision support.

6. Scope of the Project

Included

- Crime data loading and client-side processing.
- Data validation and safe handling of missing or malformed values.
- Time, city, crime-domain, category, weapon and victim analysis.
- Executive KPI cards and interactive visualizations.
- Crime trend and city comparison views.
- Case-resolution and demographic analysis.
- City-risk-oriented analytical views.
- Dynamic insights that respond to filters.

Outside the current verified scope

- Live police or emergency-system ingestion.
- Verified real-time crime feeds.
- Response-time measurement without a reliable response dataset.
- Predictive crime modeling without suitable labeled historical data.
- Production authentication, authorization and role-based security.

7. Data Source and Data Understanding

The supplied crime dataset is record-oriented. Example records contain a report number, reported date, occurrence date and time, city, crime code, crime description, victim age, victim gender, weapon used, crime domain, police deployed, case-closed status and case-closure date.

Field	Analytical meaning
Report Number	Unique report/case identifier.
Date Reported	When the report was recorded.
Date of Occurrence	Date associated with the crime occurrence.
Time of Occurrence	Time used for temporal analysis.
City	Geographic grouping for comparison.
Crime Code	Numeric crime identifier.
Crime Description	Specific reported crime type.
Victim Age / Gender	Demographic analysis.
Weapon Used	Weapon/tool associated with the record.
Crime Domain	High-level crime grouping.
Police Deployed	Recorded police deployment.
Case Closed	Case resolution status.
Date Case Closed	Closure date when available.

The source should be interpreted as reported crime records. A reported record is not automatically a complete measure of all crime, and analytical results depend on the availability and quality of the underlying fields.

8. Data Preparation and Cleaning

Cleaning objectives

- Parse and standardize reported, occurrence and closure dates.
- Extract month, year and time-of-day attributes.
- Handle missing victim age, gender and police-deployment values safely.
- Normalize categorical labels so equivalent values are grouped consistently.
- Calculate closure duration only where closure information is available.
- Prevent malformed records from crashing the dashboard.

Time-of-day classification

Period	Time range
Morning	06:00–11:59
Afternoon	12:00–16:59
Evening	17:00–20:59
Night	21:00–05:59

The preparation layer is important because downstream KPIs, filters, rankings and charts depend on consistent dates, categories, demographic values and case-status handling.

9. System Architecture and Workflow

The project follows a sequential analytical pipeline in which each stage produces an output consumed by the next stage.

Stage	Process	Output
1	Source Data	Crime dataset / JSON records
2	Data Cleaning	Validated and normalized values
3	Transformation	Date/time attributes and classifications
4	Structured Data	Fact and dimension-style analytical model
5	SQL / Python Analytics	Counts, trends and distributions
6	Risk Analysis	Derived location-oriented indicators
7	Dashboard	KPIs, charts, filters and insights

Data Flow

Crime Dataset → Cleaning → Transformation → Structured Data → SQL/Python Analytics → Risk Analysis → Dashboard

RAW DATA → CLEAN → TRANSFORM → ANALYZE → DASHBOARD

10. Data Model and Database Design

The proposed relational design uses FACT_CRIME as the central event-level table and reusable dimension tables for date, city, crime domain, crime category, weapon and victim. The structure supports grouping and filtering while reducing duplication.

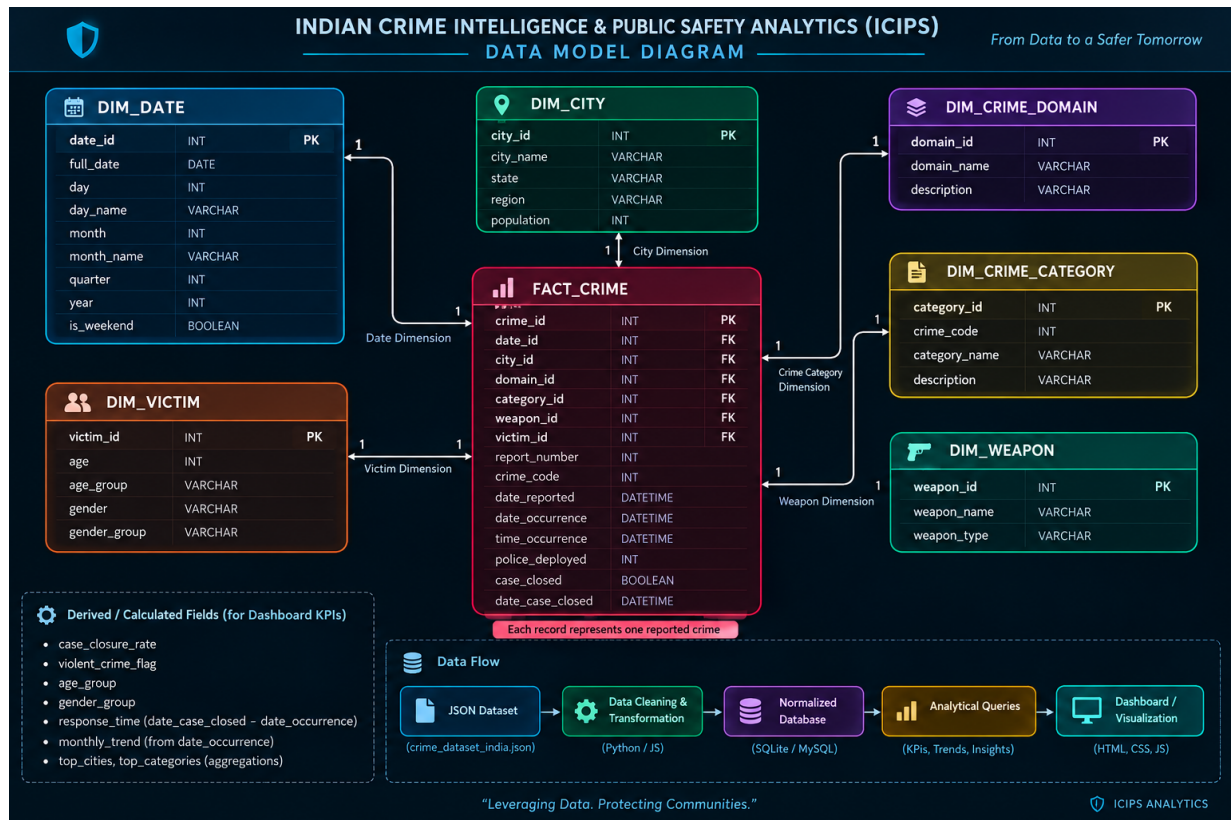


Figure 1. ICIPS crime intelligence data model supplied with the project.

Table	Key fields / role
FACT_CRIME	crime_id, date_id, city_id, domain_id, category_id, weapon_id, victim_id, report_number, dates, deployment and closure status
DIM_DATE	date_id, full_date, day, day_name, month, month_name, quarter, year, weekend flag
DIM_CITY	city_id, city_name, state, region, population
DIM_CRIME_DOMAIN	domain_id, domain_name, description
DIM_CRIME_CATEGORY	category_id, crime_code, category_name, description
DIM_WEAPON	weapon_id, weapon_name, weapon_type
DIM_VICTIM	victim_id, age, age_group, gender, gender_group

11. Analytics and KPI Calculations

The dashboard converts record-level fields into measures that can be monitored quickly and recalculated for selected filters.

Metric	Calculation / interpretation
Total Crimes	Count of report numbers within the selected scope.
Closed Cases	Count where Case Closed = Yes.
Open Cases	Count where Case Closed = No.
Case Closure Rate	$(\text{Closed Cases} / \text{Total Cases}) \times 100$.
Violent Crime %	Violent-domain records divided by total crimes $\times 100$.
Average Police Deployed	Arithmetic mean of the deployment field.
Resolution Time	Date Case Closed – Date Reported for closed cases.
Most Common Crime	Statistical mode of Crime Description.
Most Common Weapon	Statistical mode of Weapon Used.
City Ranking	Crime count aggregated by city and sorted descending.
Category Ranking	Crime count aggregated by category/description.

Analytical design principle

Every insight should remain traceable to the underlying data and the documented calculation method. Derived metrics should remain clearly distinguishable from directly measured source fields.

12. Dashboard Design and Modules

The supplied interface is a premium dark-themed, single-page analytical application with a landing page, sticky navigation and five dashboard tabs.



Figure 2. Executive dashboard view supplied with the project.

Module	Purpose / main visuals
Executive Overview	Total crime KPIs, case outcomes, trends, city rankings, domain distribution and summary insights.
Police Operations	Operational workload, deployment, backlog and city/category comparisons.
Crime Intelligence	Crime types, temporal frequencies, weapons, victim age and analytical relationships.
Women's Safety	Female-victim-oriented demographic and case-outcome analysis.
City Risk	City rankings, domain composition, resolution status, top crime types, demographics and trend timeline.

Design characteristics

- Deep navy/charcoal theme with cyan, blue, teal, orange, red and purple accents.
- Glassmorphic panels and interactive hover effects.
- Responsive layouts using Flexbox and CSS Grid.
- ApexCharts for interactive SVG visualization.
- Outfit and Caveat typography.
- Loading/progress and dataset error states.
- Filter-driven recalculation of charts and insights.

13. Dashboard Visual Analysis

The supplied project visuals show the dashboard's analytical presentation across trend, city/domain, category/demographic and insight-oriented views. The figures below are included as evidence of the implemented visual presentation described in the project materials.



Figure 3. Reported crime trend analysis view.

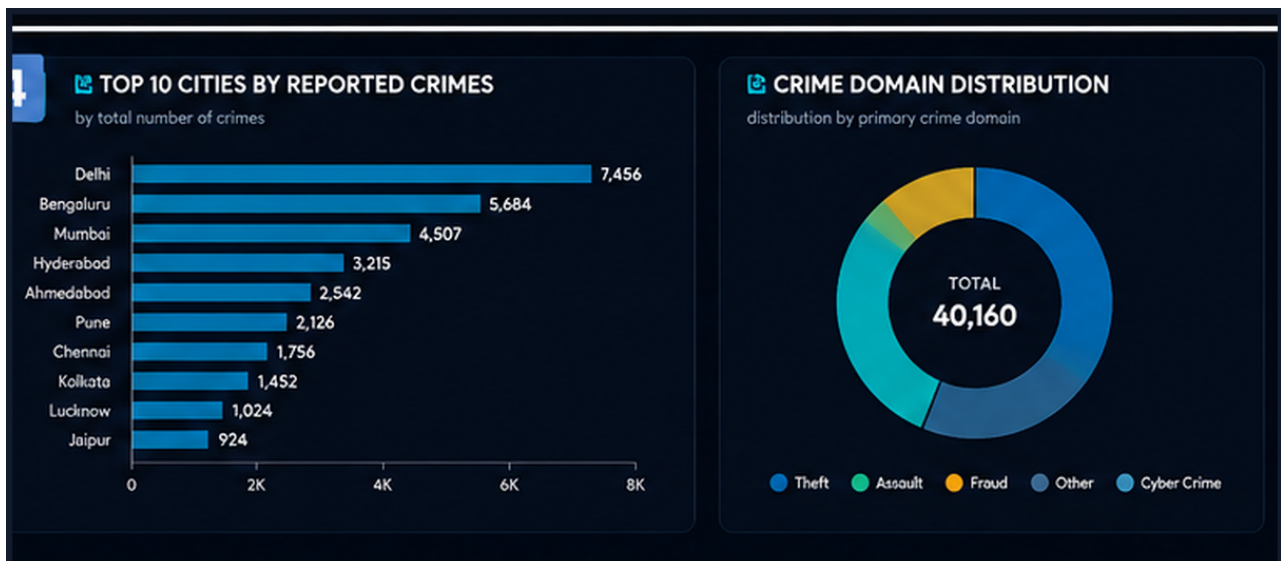


Figure 4. City and crime-domain analysis view.

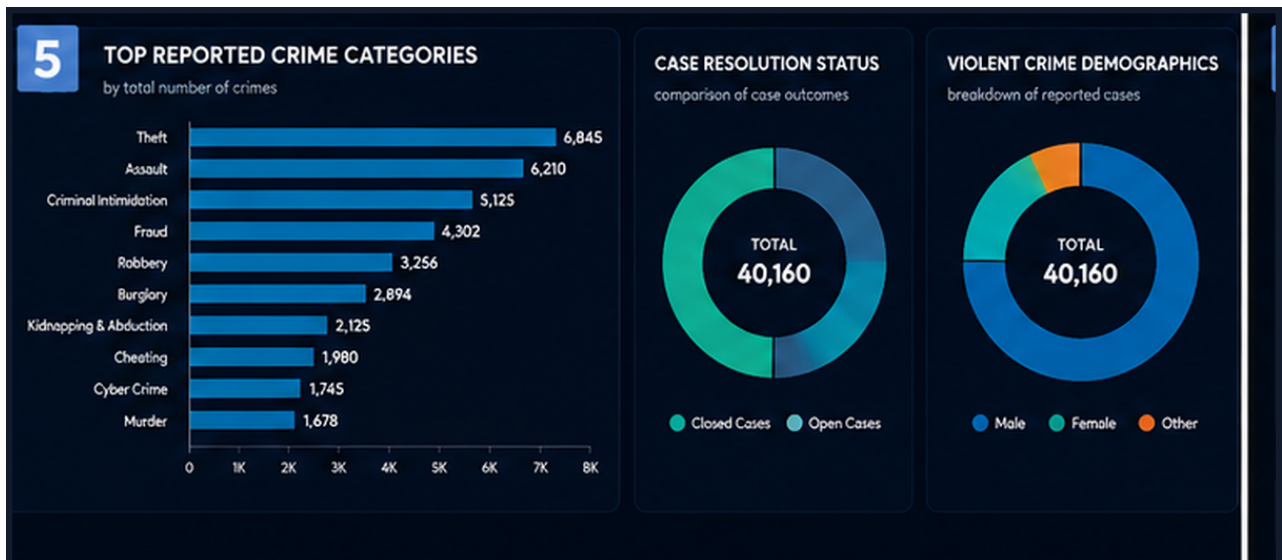


Figure 5. Crime categories and demographic analysis view.

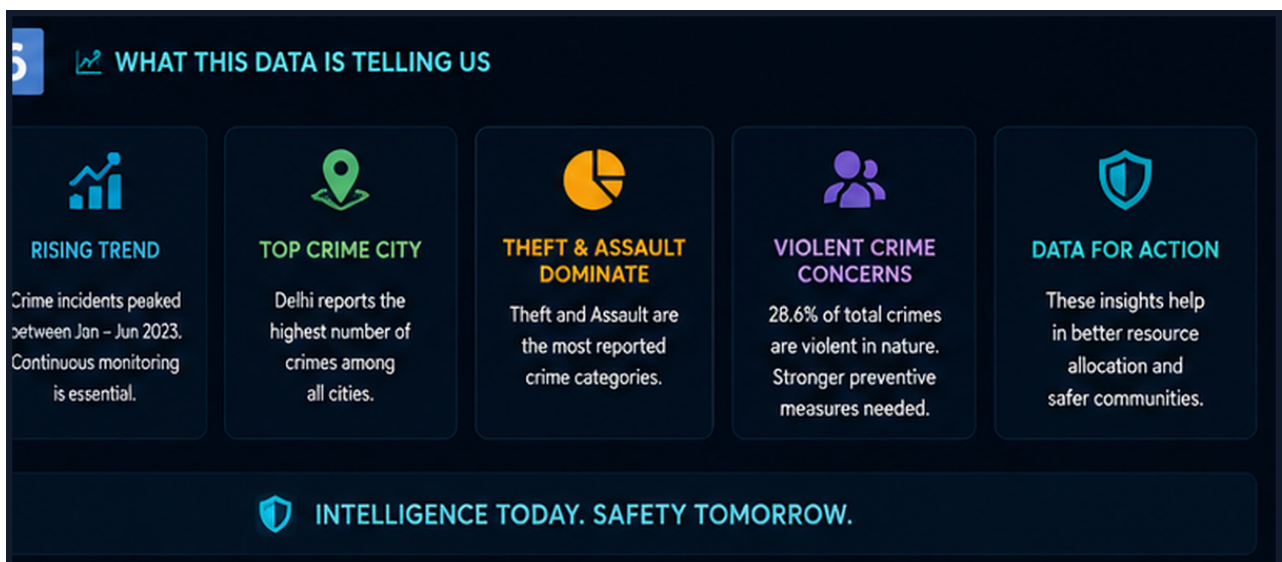


Figure 6. Key insights view.

14. Risk Analysis

Risk analysis is treated as a derived layer rather than a direct source field. A location-level risk indicator can combine measurable components such as incident frequency, severity and recency. If exposure data becomes available, incidents can also be normalized against traffic or population exposure.

Indicator	Illustrative contribution
Incident Frequency	More incidents at a location increase risk.
Severity	Severe/critical incidents receive greater weight.
Recency	Recent incidents can receive higher weight.
Exposure	Incidents can be normalized by an appropriate exposure measure when available.
Location Clustering	Repeated nearby incidents strengthen hotspot signals.

Responsible interpretation

Risk scores should not be presented as verified operational measurements unless their inputs and calculation method are validated against authoritative source data. This distinction is important because a derived risk indicator is an analytical construct rather than a directly observed source field.

15. Implementation and Technology Stack

Technology	Role
HTML5	Semantic structure and dashboard layout.
CSS3	Dark theme, responsive layout, glassmorphism and visual effects.
Vanilla JavaScript ES6+	Data loading, parsing, transformation, calculations, filters and UI control.
ApexCharts	Interactive charts and responsive visualization.
Google Fonts	Outfit and Caveat typography.
FontAwesome	Navigation and analytical icons.
Python	Cleaning, transformation and extended analytics layer.
SQL / SQLite	Normalized relational storage and analytical querying layer.

Current web application

- The current dashboard is client-side and does not require a backend framework.
- A large dataset can be embedded in a JavaScript asset for local execution.
- The application can be opened directly or served with a local static server.
- The UI includes loading and error handling for dataset initialization.

The implementation therefore supports a lightweight local demonstration while leaving a clear path toward a more structured production architecture in future work.

16. Testing and Validation

Testing focuses on data initialization, transformation, analytical correctness, filter behavior and dashboard rendering. The expected outcomes below follow the supplied project testing strategy.

Test area	Expected result
Dataset Loading	Dataset initializes without application crash.
Date Parsing	Dates and times are interpreted consistently.
Missing Values	Invalid or blank values are safely handled.
Classification	Known descriptions map to expected groups.
KPI Calculation	Totals and percentages update correctly.
Filter Behavior	Filters update dependent charts and insights.
Case Resolution	Closed/open counts and closure rate remain consistent.
City Ranking	Ordering reflects aggregated crime counts.
Dashboard Rendering	Charts and cards render at supported sizes.
Empty State	No-data selections show safe fallback messaging.

Validation principle

The application should fail safely when source values are missing or malformed, and calculated values should remain consistent with the selected analytical scope.

17. Results and Key Insights

The supplied dashboard screenshots demonstrate the executive analytical presentation. The following figures are dashboard-displayed values from the supplied project visual and are not independently revalidated source statistics.

Displayed indicator	Dashboard value
Total Crimes Reported	40,160
Closed Cases	20,062
Open Cases	20,098
Case Closure Rate	50.0%
Violent Crime Share	28.6%
Top Crime City	Delhi
Delhi – Reported Crimes	7,456
Bengaluru – Reported Crimes	5,684
Mumbai – Reported Crimes	4,507
Top Crime Category	Theft – 6,845
Second Category	Assault – 6,210
Third Category	Criminal Intimidation – 5,125

Interpretation

- The displayed executive view shows a near-even split between closed and open cases.
- Delhi is shown as the highest-volume city in the displayed top-city comparison.
- Theft and assault are prominent reported categories in the displayed ranking.
- The dashboard combines volume, resolution, domain and demographic views rather than relying on a single KPI.
- The dynamic insight area converts analytical outputs into concise statements for decision support.

18. Limitations and Data Considerations

- The source dataset represents reported crime records and should not automatically be interpreted as a complete measure of all crime.
- A metric can be calculated only when its required source fields are available.
- Real-time claims require a connected live ingestion source.
- Response-time analysis requires reliable response/dispatch timestamps.
- Weather, exposure and other contextual analysis require additional datasets.
- Severity and risk classifications are derived unless verified source fields exist.
- Missing or inconsistent source values can affect geographic and demographic analysis.
- Client-side embedding of a large dataset can affect initial loading and browser memory usage.

Data governance note

The project should preserve a clear distinction between source observations, transformed fields and derived indicators. This makes the dashboard easier to audit and prevents analytical assumptions from being mistaken for directly measured facts.

19. Future Enhancements

- Integrate live crime and public-safety data feeds.
- Add scheduled ETL jobs and automatic refresh.
- Connect police deployment and emergency dispatch systems.
- Integrate verified response-time measurements.
- Add weather and environmental context.
- Introduce automated geographic clustering and hotspot detection.
- Add authentication and role-based dashboard access.
- Deploy the dashboard as a production web application.
- Introduce predictive modeling after sufficient labeled historical data is available.
- Add exportable reports and scheduled executive summaries.

Long-term vision

The long-term goal is a validated public-safety intelligence platform that unifies historical crime records with current operational signals and presents transparent, explainable analytics to different stakeholder groups.

20. Conclusion

Indian Crime Intelligence & Public Safety Analytics demonstrates how a large crime dataset can be transformed into an interactive analytical product. The project combines data understanding, cleaning and classification with statistical calculations, visual analytics and stakeholder-oriented dashboard design.

The current dashboard communicates crime volume, case outcomes, city rankings, crime domains, categories, demographics and time-based patterns. Its client-side architecture makes the application easy to run locally and suitable for portfolio demonstration.

The project establishes a clear path toward a more rigorous platform: normalize the data, validate derived metrics, integrate additional operational sources, automate refreshes and only then introduce predictive or real-time capabilities. Every insight should remain traceable to the underlying data and documented calculation method.

INTELLIGENCE TODAY. SAFETY TOMORROW.

A. Repository Structure

Recommended repository layout:

```
ICIPS-Analytics/  
├──  
│   ├── data/  
│   │   ├── raw/  
│   │   │   ├── crime_dataset_india.json  
│   │   │   └── processed/  
│   │   └── database/  
│   │       ├── icips_analytics.db  
│   │       └── python/  
│   │           ├── data_cleaning.py  
│   │           ├── database.py  
│   │           ├── analytics.py  
│   │           └── risk_analysis.py  
│   │       └── sql/  
│   │           ├── schema.sql  
│   │           ├── analysis_queries.sql  
│   │           └── views.sql  
│   ├── dashboard/  
│   │   ├── frontend/  
│   │   │   ├── index.html  
│   │   │   ├── style.css  
│   │   │   ├── script.js  
│   │   │   └── crime_dataset.js  
│   │   └── reports/  
│   │       ├── ICIPS_Analytics_Project_Report.pdf  
│   │       └── Documentation.docx  
│   ├── images/  
│   │   ├── dashboard.png  
│   │   └── data_model.png  
│   ├── .gitignore  
│   ├── requirements.txt  
│   └── README.md
```

B. Key Definitions

Term	Definition
Crime Record	One reported crime/case record in the source dataset.
KPI	Key Performance Indicator used for quick monitoring.
Fact Table	Central event-level table used for analytical records.
Dimension Table	Reusable descriptive table used to group or filter fact records.
ETL	Extract, Transform and Load process used to prepare analytical data.
Hotspot	Geographic area showing relatively concentrated incident activity.
Risk Score	Derived indicator calculated from selected measurable factors.
Case Closure Rate	Percentage of reported cases marked closed within the selected scope.

Source basis

This report is based on the supplied ICIPS project README, project documentation, dashboard HTML/CSS/JavaScript materials, crime dataset structure, dashboard visuals and data-model information provided with the project.

C. Analytical Formula Reference

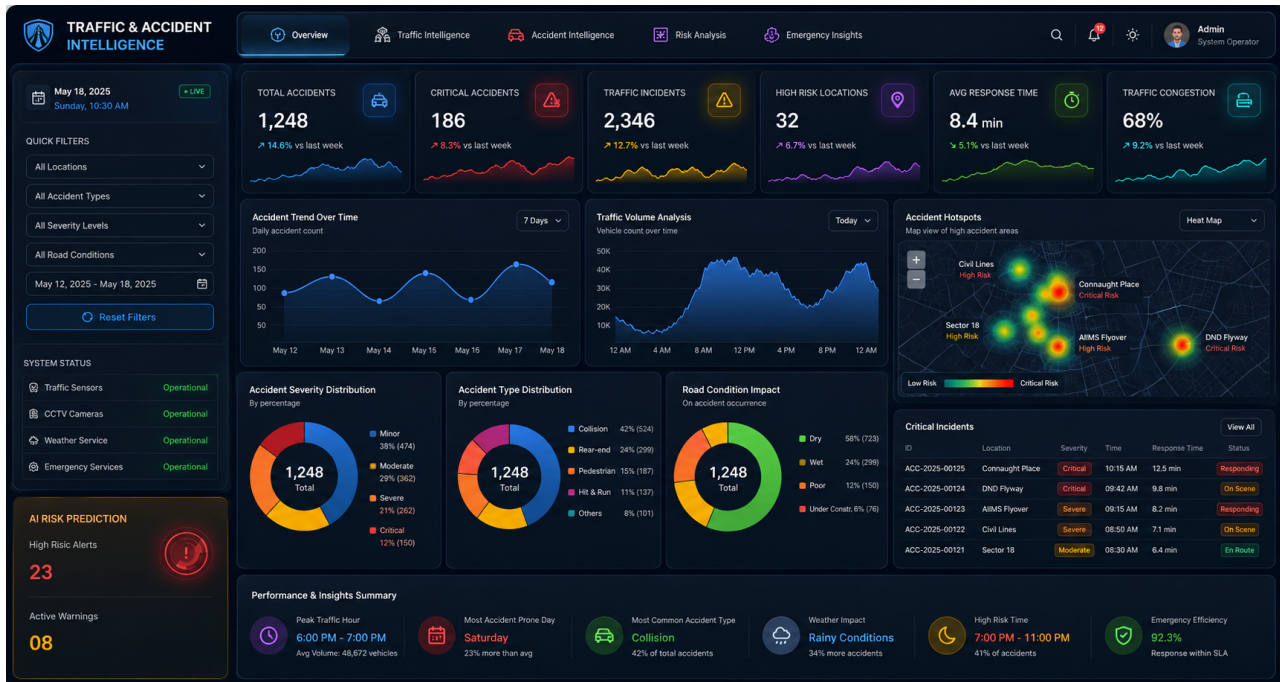
Metric	Formula / rule	Purpose
Total Crimes	COUNT(report_number)	Measure selected crime volume
Closed Cases	COUNT(Case Closed = Yes)	Measure resolved cases
Open Cases	COUNT(Case Closed = No)	Measure unresolved cases
Closure Rate	Closed / Total × 100	Measure case-resolution share
Violent Crime %	Violent records / Total × 100	Measure violent-crime share
Resolution Time	Date Closed – Date Reported	Measure elapsed closure period
City Ranking	GROUP BY City; sort count DESC	Compare reported volume by city
Category Ranking	GROUP BY Category/Description; sort DESC	Identify dominant categories
Risk Indicator	Frequency + Severity + Recency + Exposure/Clustering when available	Derived location-oriented signal

Interpretation rule

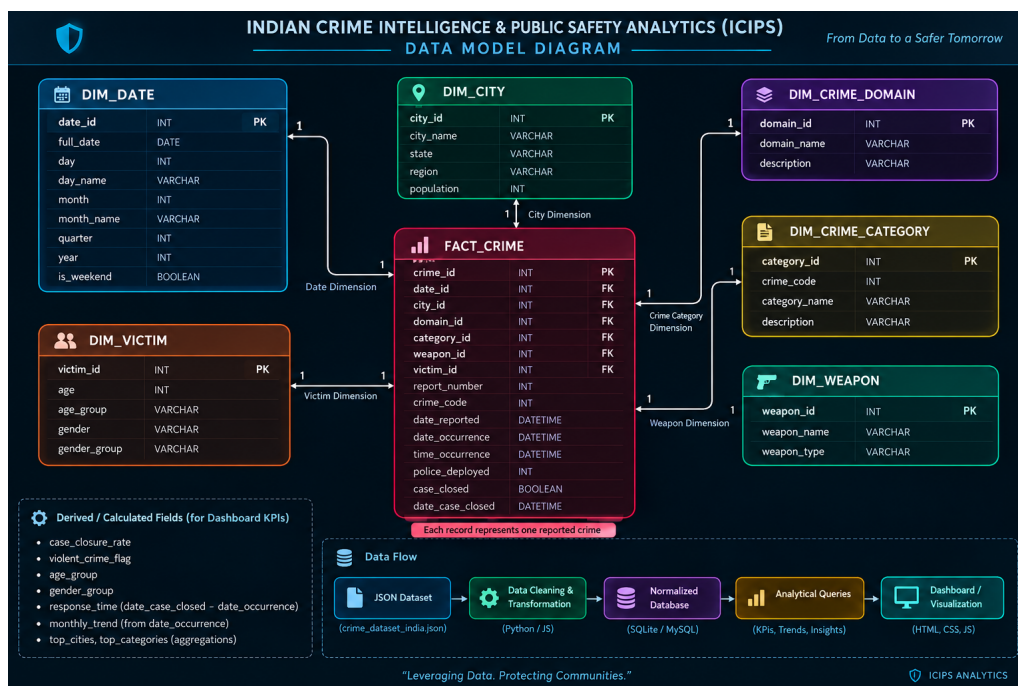
A formula is meaningful only when the underlying fields are present, valid and interpreted consistently. Risk indicators require particular care because they are derived measures rather than direct source fields.

D. Visual Evidence

The following supplied visuals document the dashboard presentation and analytical outputs included in the project.



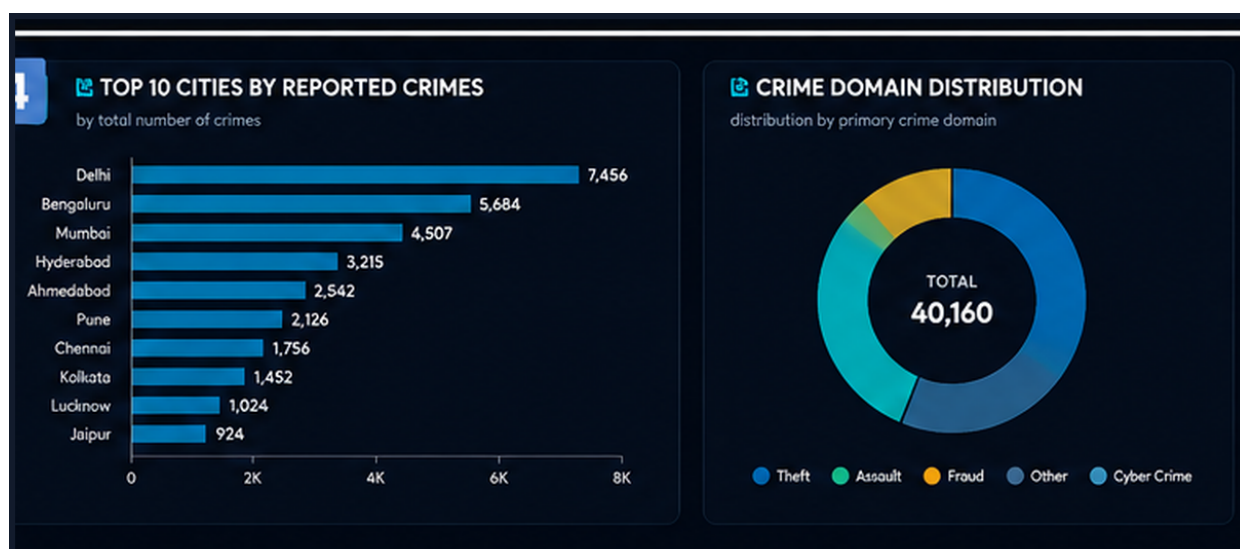
Executive dashboard



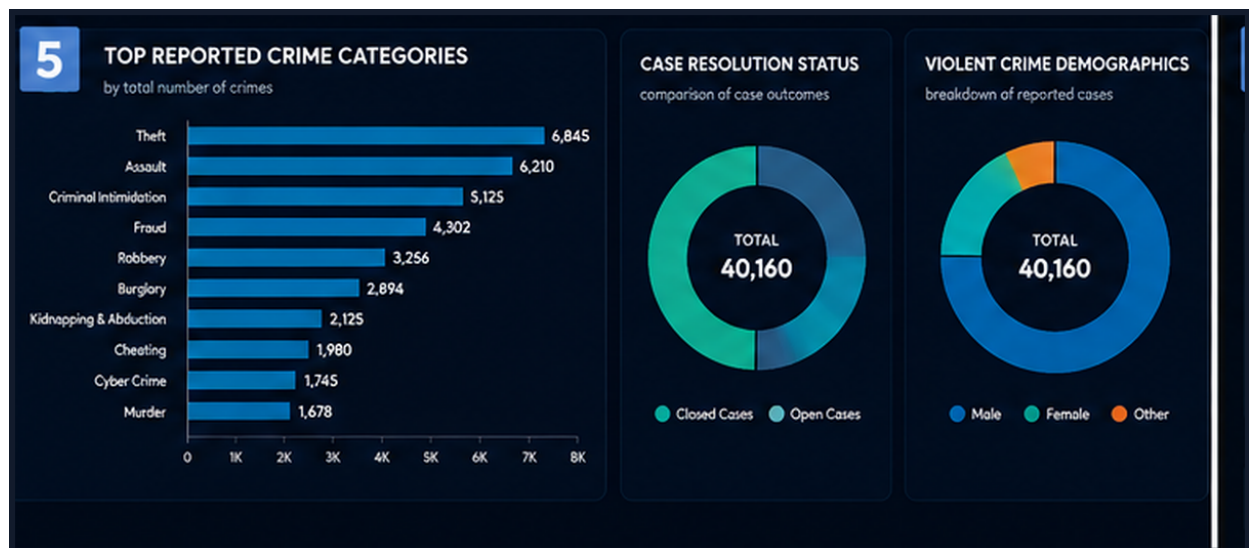
Crime intelligence data model



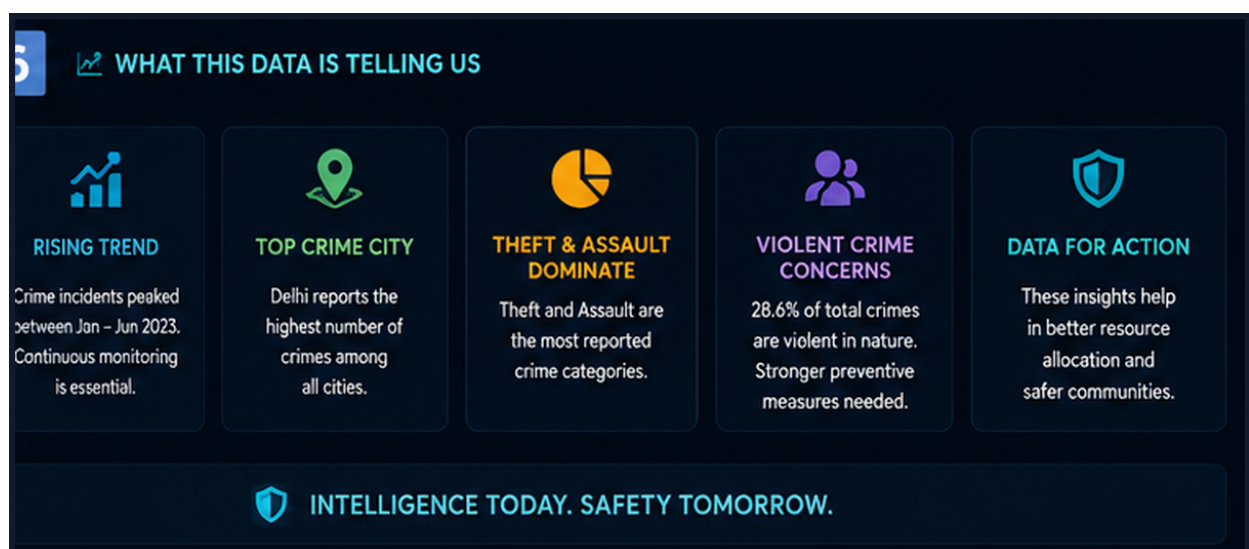
Crime trend analysis



City and domain analysis



Crime categories and demographics



Key insights